🚢 Titanic Dataset — End-to-End Exploratory Data Analysis

Project 1 | SyntecxHub Data Science Series

Dataset Source: [Kaggle Titanic Competition]

📋 Project Objectives

1. Load the Titanic dataset, inspect missingness and data types.
2. Analyze survival rates by **Sex**, **Passenger Class**, and **Age buckets**.
3. Visualize findings with bar charts, violin plots, and heatmaps.
4. Generate a short insight report with key findings.

📦 Step 1 — Install & Import Libraries

```
# Install missing libraries (Colab usually has these, but just in case)
!pip install pandas matplotlib seaborn missingno -q

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
import missingno as msno
import warnings

warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid', palette='muted')
plt.rcParams.update({'figure.dpi': 120, 'font.family': 'DejaVu Sans'})

print('Libraries imported successfully!')
```

```
Libraries imported successfully!
```

📂 Step 2 — Load Dataset

> **How to get the data:**

- Option A: Download from [Kaggle Titanic]() and upload `train.csv` to Colab.
- Option B: Use the Kaggle API (cell below).

```
#Upload manually ————————————————————————————————————————
from google.colab import files
uploaded = files.upload()  # Upload train.csv
df = pd.read_csv('train.csv')

print(f'Dataset loaded: {df.shape[0]} rows × {df.shape[1]} columns')
df.head()
```

Choose Files 3 files

gender_submission.csv(text/csv) - 3258 bytes, last modified: 4/21/2026 - 100% done
test.csv(text/csv) - 28629 bytes, last modified: 4/21/2026 - 100% done
train.csv(text/csv) - 61194 bytes, last modified: 4/21/2026 - 100% done

```
Saving gender_submission.csv to gender_submission.csv
Saving test.csv to test.csv
Saving train.csv to train.csv
Dataset loaded: 891 rows × 12 columns
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 2117
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs	female	38.0	1	0	PC 1759

Next steps: Generate code with df | New interactive sheet

🔍 Step 3 — Initial Inspection (Dtypes & Missingness)

```
print('=' * 50)
print('DATASET INFO')
print('=' * 50)
df.info()

print('\n' + '=' * 50)
print('DESCRIPTIVE STATISTICS')
print('=' * 50)
df.describe(include='all').T
```

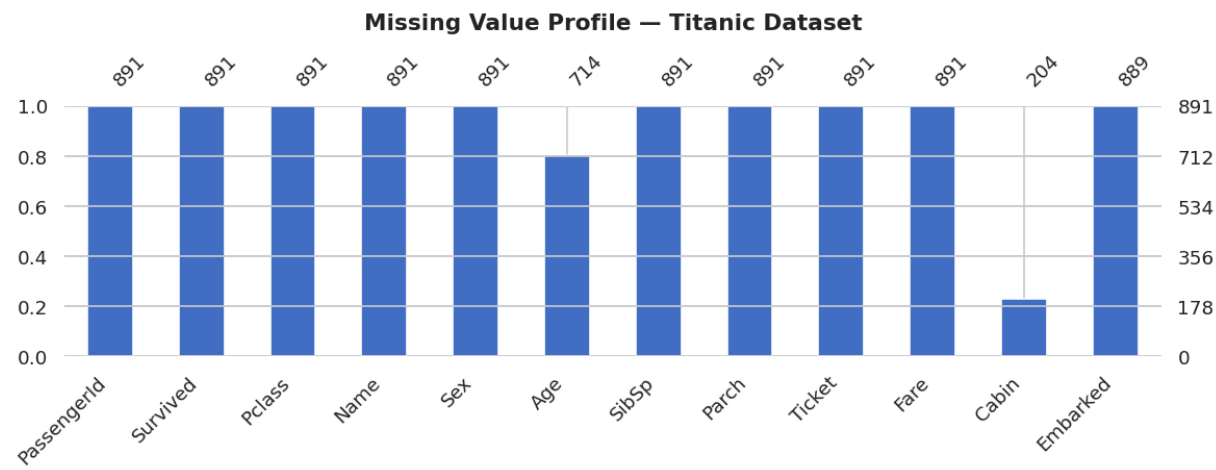
```
==================================================
DATASET INFO
==================================================
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

==================================================
DESCRIPTIVE STATISTICS
==================================================
```

	count	unique	top	freq	mean	std	min	25%	
PassengerId	891.0	NaN	NaN	NaN	446.0	257.353842	1.0	223.5	
Survived	891.0	NaN	NaN	NaN	0.383838	0.486592	0.0	0.0	
Pclass	891.0	NaN	NaN	NaN	2.308642	0.836071	1.0	2.0	
Name	891	891	Dooley, Mr. Patrick	1	NaN	NaN	NaN	NaN	
Sex	891	2	male	577	NaN	NaN	NaN	NaN	
Age	714.0	NaN	NaN	NaN	29.699118	14.526497	0.42	20.125	
SibSp	891.0	NaN	NaN	NaN	0.523008	1.102743	0.0	0.0	
Parch	891.0	NaN	NaN	NaN	0.381594	0.806057	0.0	0.0	
Ticket	891	681	347082	7	NaN	NaN	NaN	NaN	
Fare	891.0	NaN	NaN	NaN	32.204208	49.693429	0.0	7.9104	14
Cabin	204	147	G6	4	NaN	NaN	NaN	NaN	

```
# Missing value summary
missing = df.isnull().sum().sort_values(ascending=False)
missing_pct = (missing / len(df) * 100).round(2)
```

```
miss_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct})
miss_df = miss_df[miss_df['Missing Count'] > 0]

print('\n📊 Missing Value Report:')
print(miss_df.to_string())

# Visual: Missing value matrix
fig, ax = plt.subplots(figsize=(10, 4))
msno.bar(df, ax=ax, color='#4472C4', fontsize=11)
ax.set_title('Missing Value Profile — Titanic Dataset', fontsize=13, fontweigl
plt.tight_layout()
plt.savefig('missing_values.png', bbox_inches='tight')
plt.show()
print('\n⚠️  Cabin (77%) and Age (20%) have significant missingness — handle
```

```
📊 Missing Value Report:
          Missing Count  Missing %
Cabin               687      77.10
Age                 177      19.87
Embarked              2       0.22
```




```
⚠️  Cabin (77%) and Age (20%) have significant missingness — handle before mod
```

🧹 Step 4 — Data Cleaning & Feature Engineering

```python
df_clean = df.copy()

# 1. Fill missing Age with median by Pclass & Sex
df_clean['Age'] = df_clean.groupby(['Pclass', 'Sex'])['Age'].transform(
    lambda x: x.fillna(x.median())
)

# 2. Fill Embarked with mode
df_clean['Embarked'].fillna(df_clean['Embarked'].mode()[0], inplace=True)

# 3. Create AgeBucket feature
df_clean['AgeBucket'] = pd.cut(
    df_clean['Age'],
    bins=[0, 12, 18, 35, 60, 100],
    labels=['Child (0-12)', 'Teen (13-18)', 'Adult (19-35)', 'Middle (36-60)',
)

# 4. Create FareGroup
df_clean['FareGroup'] = pd.qcut(
    df_clean['Fare'], q=4,
    labels=['Low', 'Medium', 'High', 'Very High']
)

# 5. Family size
df_clean['FamilySize'] = df_clean['SibSp'] + df_clean['Parch'] + 1
df_clean['IsAlone'] = (df_clean['FamilySize'] == 1).astype(int)

print('✅ Feature engineering complete!')
print(f"Age nulls remaining: {df_clean['Age'].isnull().sum()}")
df_clean[['Age', 'AgeBucket', 'FamilySize', 'IsAlone', 'FareGroup']].head(8)
```

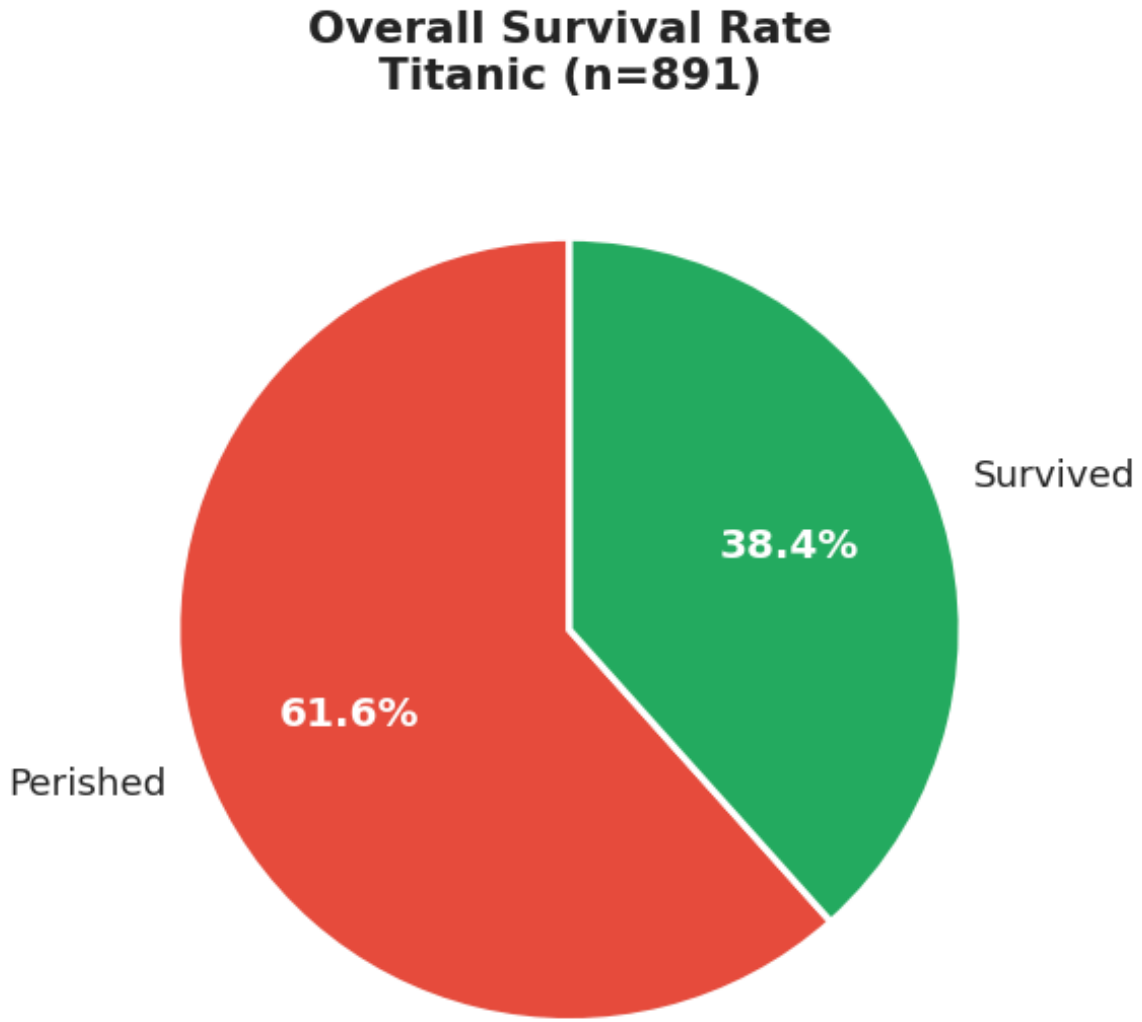
```
✅ Feature engineering complete!
Age nulls remaining: 0
```

	Age	AgeBucket	FamilySize	IsAlone	FareGroup
0	22.0	Adult (19-35)	2	0	Low
1	38.0	Middle (36-60)	2	0	Very High
2	26.0	Adult (19-35)	1	1	Medium
3	35.0	Adult (19-35)	2	0	Very High
4	35.0	Adult (19-35)	1	1	Medium
5	25.0	Adult (19-35)	1	1	Medium
6	54.0	Middle (36-60)	1	1	Very High
7	2.0	Child (0-12)	5	0	High

📊 Step 5 — Overall Survival Rate

```
survived = df_clean['Survived'].sum()
total = len(df_clean)
rate = df_clean['Survived'].mean()

print(f'Total Passengers : {total}')
print(f'Survived         : {survived} ({rate:.1%})')
print(f'Perished         : {total - survived} ({1-rate:.1%})')

fig, ax = plt.subplots(figsize=(5, 5))
colors = ['#E74C3C', '#27AE60']
wedges, texts, autotexts = ax.pie(
    [total - survived, survived],
    labels=['Perished', 'Survived'],
    autopct='%1.1f%%',
    colors=colors,
    startangle=90,
    wedgeprops={'edgecolor': 'white', 'linewidth': 2}
)
for at in autotexts:
    at.set_fontsize(12)
    at.set_fontweight('bold')
    at.set_color('white')

ax.set_title('Overall Survival Rate\nTitanic (n=891)', fontsize=13, fontweight
plt.tight_layout()
plt.savefig('survival_overall.png', bbox_inches='tight')
plt.show()
```

```
Total Passengers : 891
Survived         : 342 (38.4%)
Perished         : 549 (61.6%)
```

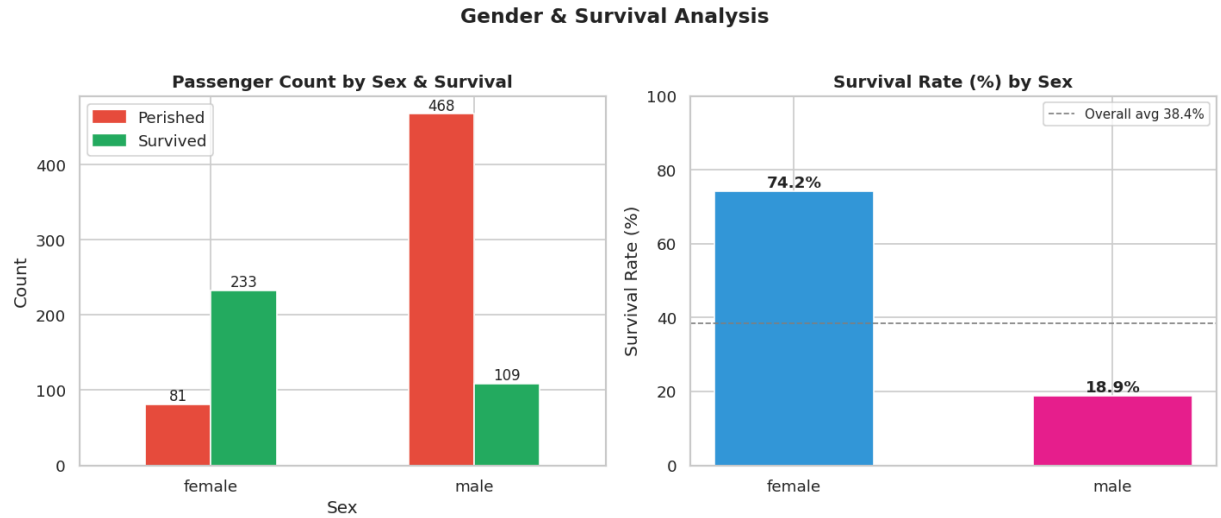


👥 Step 6 — Survival by Sex

```
sex_surv = df_clean.groupby('Sex')['Survived'].agg(['sum', 'count', 'mean']).r
sex_surv.columns = ['Sex', 'Survived', 'Total', 'Rate']
print(sex_surv.to_string(index=False))

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Plot 1: Count
ax = axes[0]
surv_count = df_clean.groupby(['Sex', 'Survived']).size().unstack()
surv_count.plot(kind='bar', ax=ax, color=['#E74C3C', '#27AE60'],
                edgecolor='white', rot=0, legend=False)
ax.set_title('Passenger Count by Sex & Survival', fontsize=12, fontweight='bo
```

```
ax.set_xlabel('Sex'); ax.set_ylabel('Count')
labels = [mpatches.Patch(color='#E74C3C', label='Perished'),
          mpatches.Patch(color='#27AE60', label='Survived')]
ax.legend(handles=labels)
for p in ax.patches:
    ax.annotate(str(int(p.get_height())),
                (p.get_x() + p.get_width()/2, p.get_height()),
                ha='center', va='bottom', fontsize=10)

# Plot 2: Rate
ax = axes[1]
colors_sex = ['#3498DB', '#E91E8C']
bars = ax.bar(sex_surv['Sex'], sex_surv['Rate']*100, color=colors_sex,
              edgecolor='white', width=0.5)
ax.set_title('Survival Rate (%) by Sex', fontsize=12, fontweight='bold')
ax.set_ylabel('Survival Rate (%)'); ax.set_ylim(0, 100)
ax.axhline(rate*100, color='gray', linestyle='--', linewidth=1, label=f'Overal
ax.legend(fontsize=9)
for bar, r in zip(bars, sex_surv['Rate']):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'{r:.1%}', ha='center', fontweight='bold', fontsize=11)

plt.suptitle('Gender & Survival Analysis', fontsize=14, fontweight='bold', y=1
plt.tight_layout()
plt.savefig('survival_by_sex.png', bbox_inches='tight')
plt.show()
print('\n🔑 Key: Females survived at 74.2% vs Males at 18.9% — "Women & child
```

```
   Sex  Survived  Total     Rate
female       233    314 0.742038
  male       109    577 0.188908
```



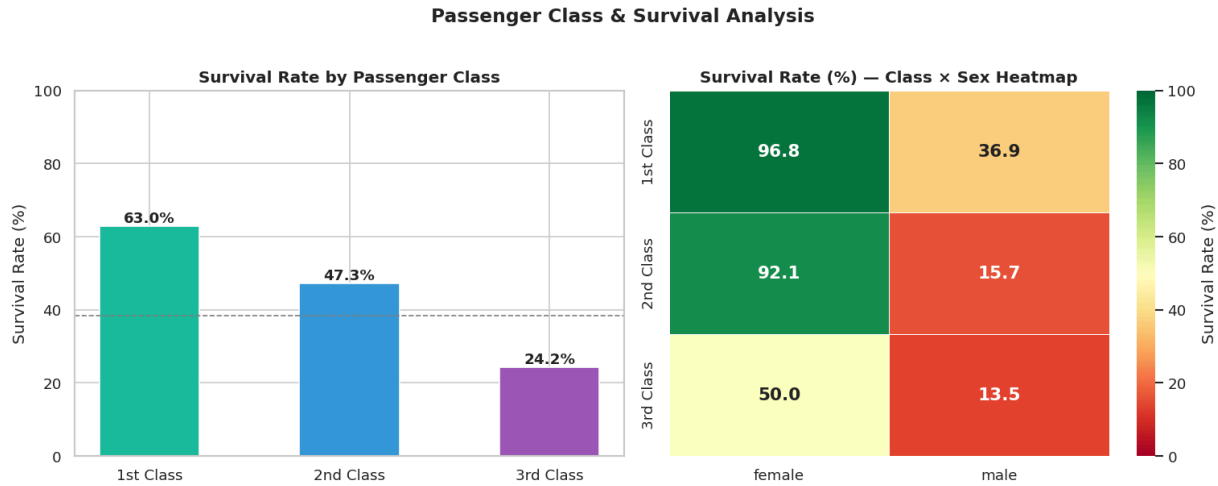
🔑 Key: Females survived at 74.2% vs Males at 18.9% — "Women & children first"

🎫 Step 7 — Survival by Passenger Class

```
pclass_surv = df_clean.groupby('Pclass')['Survived'].agg(['mean','sum','count
pclass_surv.columns = ['Pclass', 'Rate', 'Survived', 'Total']
pclass_surv['PclassLabel'] = pclass_surv['Pclass'].map({1:'1st Class', 2:'2nd
print(pclass_surv[['PclassLabel','Survived','Total','Rate']].to_string(index=

# Grouped bar by Class + Sex
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Left: Survival rate by class
ax = axes[0]
```

```
class_colors = ['#1ABC9C', '#3498DB', '#9B59B6']
bars = ax.bar(pclass_surv['PclassLabel'], pclass_surv['Rate']*100,
              color=class_colors, edgecolor='white', width=0.5)
ax.set_title('Survival Rate by Passenger Class', fontsize=12, fontweight='bol
ax.set_ylabel('Survival Rate (%)')
ax.set_ylim(0, 100)
ax.axhline(rate*100, color='gray', linestyle='--', linewidth=1)
for bar, r in zip(bars, pclass_surv['Rate']):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'{r:.1%}', ha='center', fontweight='bold', fontsize=11)

# Right: Heatmap Class x Sex
ax = axes[1]
pivot = df_clean.pivot_table('Survived', index='Pclass', columns='Sex', aggfu
pivot.index = ['1st Class', '2nd Class', '3rd Class']
sns.heatmap(pivot, annot=True, fmt='.1f', cmap='RdYlGn', ax=ax,
            linewidths=0.5, linecolor='white', cbar_kws={'label': 'Survival R
            vmin=0, vmax=100, annot_kws={'size': 13, 'weight': 'bold'})
ax.set_title('Survival Rate (%) — Class × Sex Heatmap', fontsize=12, fontweig
ax.set_xlabel(''); ax.set_ylabel('')

plt.suptitle('Passenger Class & Survival Analysis', fontsize=14, fontweight='
plt.tight_layout()
plt.savefig('survival_by_class.png', bbox_inches='tight')
plt.show()
print('\n🔑 1st class females: 96.8% survival | 3rd class males: 13.5% surviv
```

```
PclassLabel  Survived  Total     Rate
  1st Class       136    216 0.629630
  2nd Class        87    184 0.472826
  3rd Class       119    491 0.242363
```

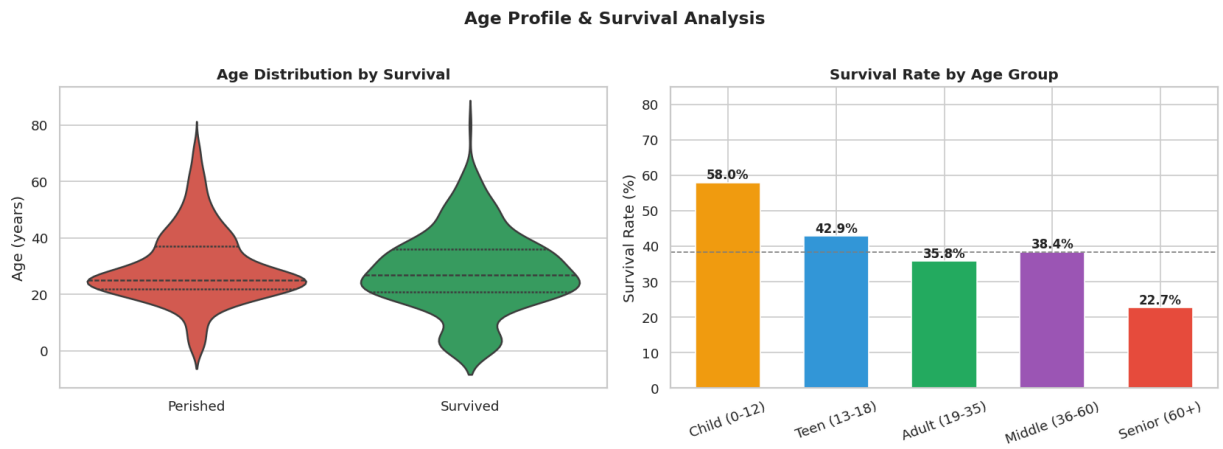


🔑 1st class females: 96.8% survival | 3rd class males: 13.5% survival

🎂 Step 8 — Survival by Age (Violin + Bar)

```
age_surv = df_clean.groupby('AgeBucket')['Survived'].mean().reset_index()

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Violin plot
ax = axes[0]
sns.violinplot(data=df_clean, x='Survived', y='Age',
               palette=['#E74C3C', '#27AE60'],
               inner='quartile', ax=ax, linewidth=1.5)
ax.set_xticks([0, 1])
ax.set_xticklabels(['Perished', 'Survived'], fontsize=11)
```

```
ax.set_title('Age Distribution by Survival', fontsize=12, fontweight='bold')
ax.set_xlabel(''); ax.set_ylabel('Age (years)')

# Bar chart by age bucket
ax = axes[1]
bucket_colors = ['#F39C12', '#3498DB', '#27AE60', '#9B59B6', '#E74C3C']
bars = ax.bar(age_surv['AgeBucket'].astype(str), age_surv['Survived']*100,
              color=bucket_colors, edgecolor='white', width=0.6)
ax.set_title('Survival Rate by Age Group', fontsize=12, fontweight='bold')
ax.set_ylabel('Survival Rate (%)')
ax.set_ylim(0, 85)
ax.tick_params(axis='x', rotation=20)
ax.axhline(rate*100, color='gray', linestyle='--', linewidth=1)
for bar, r in zip(bars, age_surv['Survived']):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'{r:.1%}', ha='center', fontweight='bold', fontsize=10)

plt.suptitle('Age Profile & Survival Analysis', fontsize=14, fontweight='bold
plt.tight_layout()
plt.savefig('survival_by_age.png', bbox_inches='tight')
plt.show()
print('\n🔑 Children had the highest survival rate (58%), Seniors the lowest
```

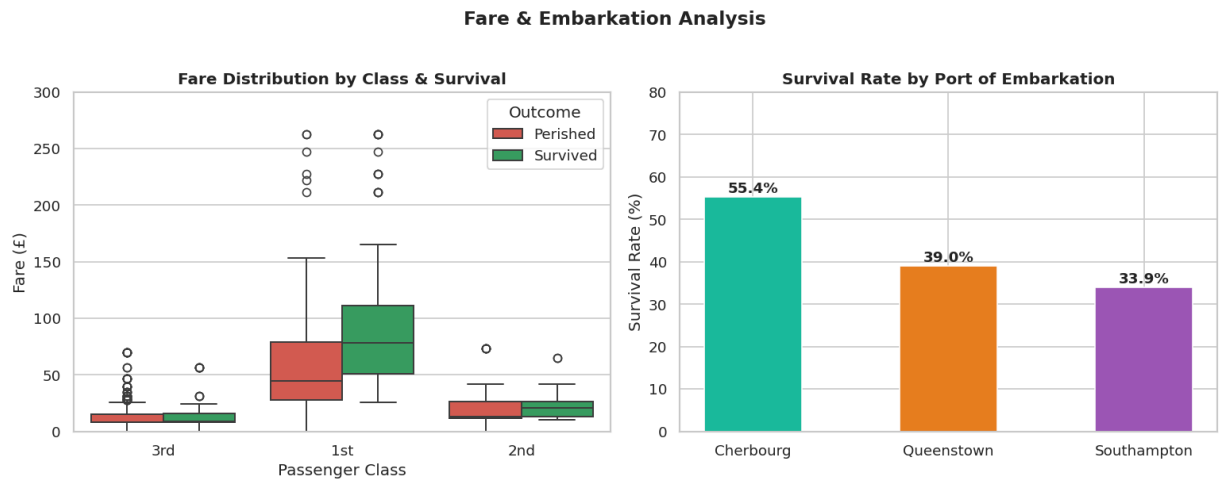


🔑 Children had the highest survival rate (58%), Seniors the lowest (22.7%)

💰 Step 9 — Fare & Embarkation Analysis

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Boxplot: Fare by class
ax = axes[0]
df_clean['PclassLabel'] = df_clean['Pclass'].map({1:'1st', 2:'2nd', 3:'3rd'})
sns.boxplot(data=df_clean, x='PclassLabel', y='Fare', hue='Survived',
            palette={0:'#E74C3C', 1:'#27AE60'}, ax=ax, linewidth=1.2)
ax.set_title('Fare Distribution by Class & Survival', fontsize=12, fontweight='
ax.set_xlabel('Passenger Class'); ax.set_ylabel('Fare (£)')
ax.set_ylim(0, 300)
handles, labels = ax.get_legend_handles_labels()
ax.legend(handles, ['Perished', 'Survived'], title='Outcome')

# Bar: Survival by Embarkation port
ax = axes[1]
emb_map = {'C': 'Cherbourg', 'Q': 'Queenstown', 'S': 'Southampton'}
df_clean['EmbarkLabel'] = df_clean['Embarked'].map(emb_map)
```

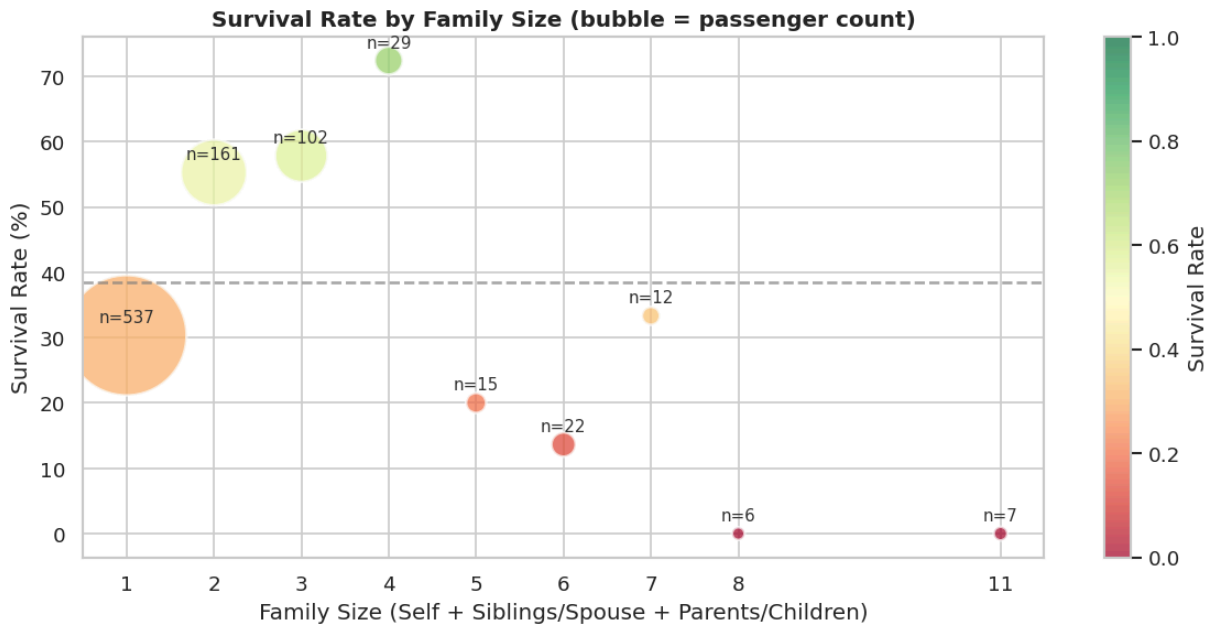
```
emb_rate = df_clean.groupby('EmbarkLabel')['Survived'].mean().reset_index()
emb_colors = ['#1ABC9C', '#E67E22', '#9B59B6']
bars = ax.bar(emb_rate['EmbarkLabel'], emb_rate['Survived']*100,
              color=emb_colors, edgecolor='white', width=0.5)
ax.set_title('Survival Rate by Port of Embarkation', fontsize=12, fontweight='b
ax.set_ylabel('Survival Rate (%)'); ax.set_ylim(0, 80)
for bar, r in zip(bars, emb_rate['Survived']):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'{r:.1%}', ha='center', fontweight='bold', fontsize=11)

plt.suptitle('Fare & Embarkation Analysis', fontsize=14, fontweight='bold', y=1
plt.tight_layout()
plt.savefig('fare_embarkation.png', bbox_inches='tight')
plt.show()
```



👪 Step 10 — Family Size Analysis

```
fig, ax = plt.subplots(figsize=(10, 5))
fam_surv = df_clean.groupby('FamilySize')['Survived'].agg(['mean','count']).re
```

```
fam_surv.columns = ['FamilySize', 'SurvivalRate', 'Count']

# Bubble chart
scatter = ax.scatter(fam_surv['FamilySize'], fam_surv['SurvivalRate']*100,
                     s=fam_surv['Count']*8, alpha=0.7,
                     c=fam_surv['SurvivalRate'], cmap='RdYlGn',
                     vmin=0, vmax=1, edgecolors='white', linewidth=1.5)

for _, row in fam_surv.iterrows():
    ax.annotate(f"n={int(row['Count'])}",
                (row['FamilySize'], row['SurvivalRate']*100 + 2),
                ha='center', fontsize=9, color='#333')

plt.colorbar(scatter, ax=ax, label='Survival Rate')
ax.set_title('Survival Rate by Family Size (bubble = passenger count)',
             fontsize=12, fontweight='bold')
ax.set_xlabel('Family Size (Self + Siblings/Spouse + Parents/Children)')
ax.set_ylabel('Survival Rate (%)')
ax.set_xticks(fam_surv['FamilySize'])
ax.axhline(rate*100, color='gray', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.savefig('family_size.png', bbox_inches='tight')
plt.show()
print('\n🔑 Solo travelers and very large families (7+) had below-average sur
```

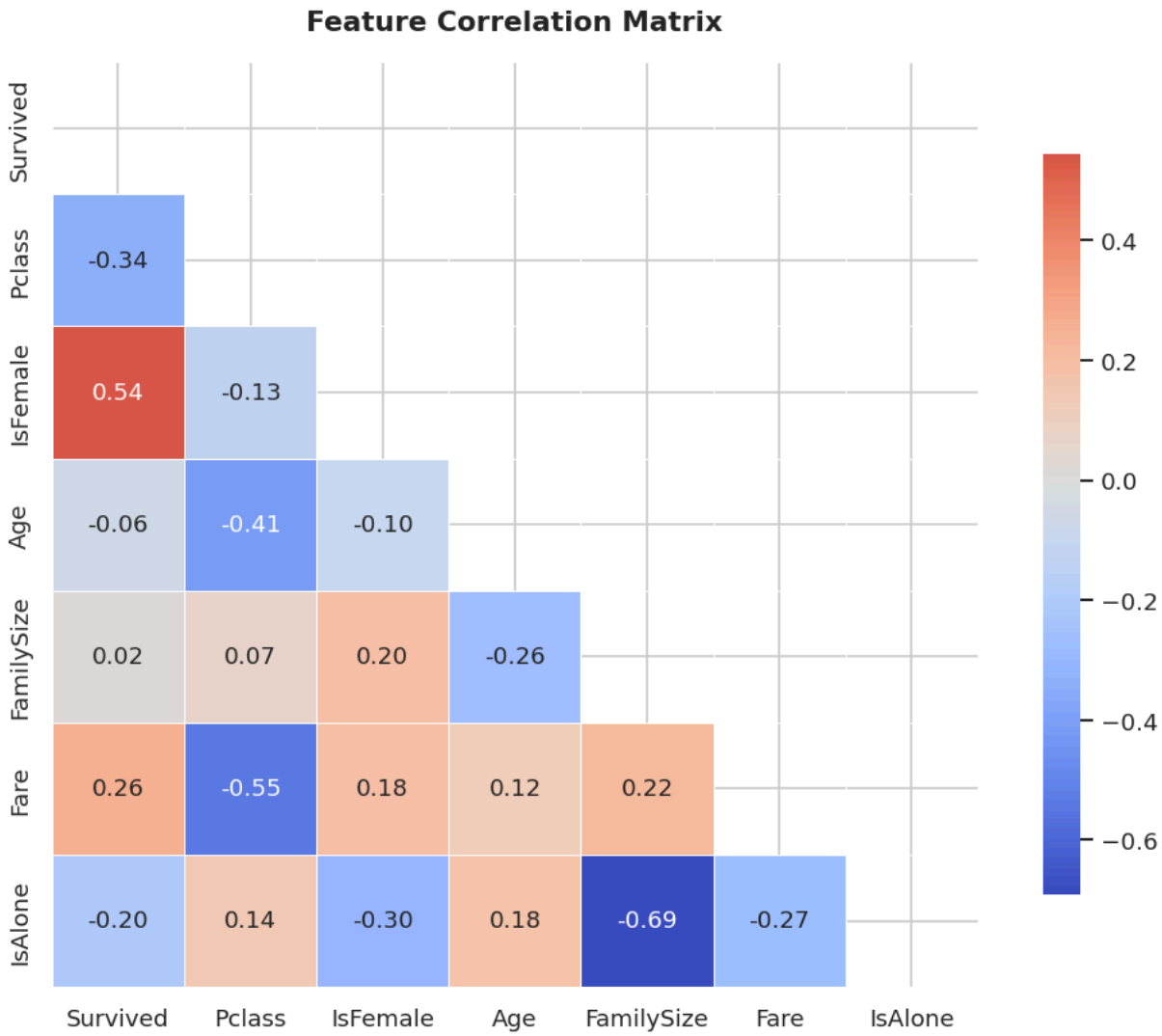


🔑 Solo travelers and very large families (7+) had below-average survival.

🔗 Step 11 — Correlation Heatmap

```
df_corr = df_clean.copy()
df_corr['IsFemale'] = (df_corr['Sex'] == 'female').astype(int)
numeric_cols = ['Survived', 'Pclass', 'IsFemale', 'Age', 'FamilySize', 'Fare',
corr_matrix = df_corr[numeric_cols].corr()

fig, ax = plt.subplots(figsize=(9, 7))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
sns.heatmap(corr_matrix, mask=mask, annot=True, fmt='.2f', cmap='coolwarm',
            ax=ax, center=0, square=True, linewidths=0.5,
            cbar_kws={'shrink': 0.8}, annot_kws={'size': 11})
ax.set_title('Feature Correlation Matrix', fontsize=13, fontweight='bold', pad=
plt.tight_layout()
plt.savefig('correlation_heatmap.png', bbox_inches='tight')
```

```
plt.show()
print('\n🔑 Strongest predictors of survival: Sex (IsFemale: +0.54), Pclass (-
```




```
🔑 Strongest predictors of survival: Sex (IsFemale: +0.54), Pclass (-0.34), Fa
```

Step 12 — Key Insight Report